



Y COMBINATOR TASK



Hi everyone, my name is **Amr Ismail**, and I am a Frontend Developer. Today, I want to walk through my implementation for this technical assessment. My main goal was to focus on the frontend architecture, clean code, and providing a top-tier user experience.

"Let's start with the Node Abstraction. When I looked at the initial code, I noticed a lot of repetition. To fix this, I created a single, reusable component called the **BaseNode**. Now, all nodes—like the Input, Output, and LLM nodes—wrap around this **BaseNode**. This makes the code very clean, easy to maintain, and highly scalable if we want to add fifty new nodes in the future."

The second big challenge was the Text Node. I completely rebuilt its logic. First, I made the text box auto-resize. As the user types more lines, the box smoothly expands downwards, but the connection handles stay perfectly fixed in place so the UI doesn't break or shake.

Finally, the styling. I didn't want it to look just like a test; I wanted it to look like a real, modern SaaS product. I unified the design system across all files so that all inputs, selects, and text areas share the same clean look. I also integrated framer-motion to add smooth hover effects and entry animations, giving the user a very premium feel.

As my primary expertise and passion lie in frontend development, I made the decision to invest my time entirely into perfecting the UI, the user experience, and the component architecture. I sincerely apologize for not completing the backend integration portion. My goal was to deliver a polished frontend product that truly reflects the quality of my work. Thank you for your time and understanding, and I would be more than happy to answer any questions.